

Robot Hardware Map

FRC Java Coding Lab 7.0 - Real Robot Programming

Updated from Phoenix Tuner X verification - 2026-07-29

1. Drivebase

Item	Verified / Current Value
Drive type	Tank Drive
Motor controllers	4 x REV SPARK MAX
Motors	4 x brushed CIM
Left leader / follower	CAN 11 -> CAN 8
Right leader / follower	CAN 10 -> CAN 7
Current limit	60 A
Voltage compensation	12 V
Encoder	Leader integrated relative encoder

2. Intake

Item	Verified / Current Value
Motor	Kraken X60
Integrated controller	CTRE Talon FX
CAN ID	1
CAN bus	rio
Phoenix 6 application version	26.2.0.0
Device state	Running Application
Firmware status	Update available
Software API	CTRE Phoenix 6 TalonFX

3. Shooter Flywheel

Item	Verified / Current Value
Motor	Kraken X60
Integrated controller	CTRE Talon FX
CAN ID	9
CAN bus	rio
Phoenix 6 application version	26.2.0.0
Device state	Running Application
Firmware status	Update available
Software API	CTRE Phoenix 6 TalonFX

4. External Sensor

Item	Verified / Current Value
Device	CTRE CANcoder
CAN ID	4

CAN bus	rio
Phoenix 6 application version	26.1.0.0
Device state	Running Application
Software API	CTRE Phoenix 6 CANcoder

5. Loader / Additional Mechanism

Item	Verified / Current Value
Status	Not confirmed in the supplied Phoenix Tuner X detection view
Previous map entry	REV SPARK MAX, CAN 19
Authority	Do not treat CAN 19 as verified until the physical device is detected and identified

6. Verified CAN Device Summary

Function	Device	Vendor	CAN ID	CAN Bus
Intake	Kraken X60 / Talon FX	CTRE	1	rio
Shooter Flywheel	Kraken X60 / Talon FX	CTRE	9	rio
External Sensor	CANcoder	CTRE	4	rio
Drive Left Leader	SPARK MAX	REV	11	rio
Drive Left Follower	SPARK MAX	REV	8	rio
Drive Right Leader	SPARK MAX	REV	10	rio
Drive Right Follower	SPARK MAX	REV	7	rio

Implementation authority: Phoenix Tuner X detection is authoritative for the CTRE devices shown above. The legacy hardware map entries that identified the Intake and Shooter Flywheel as REV SPARK MAX devices are superseded.

Required code mapping: Intake must use CTRE Phoenix 6 TalonFX on CAN ID 1. Shooter Flywheel must use CTRE Phoenix 6 TalonFX on CAN ID 9. Do not instantiate REV SparkMax objects for these two mechanisms.